

Scalability issues: Centralised platforms can become overwhelmed as the number of connected devices increases, leading to latency and performance issues.

In contrast, decentralised IoT systems eliminate the need for centralised control, using blockchain and peer-to-peer communication to securely and efficiently exchange data across devices without intermediaries. This offers several benefits.

Improved security: By distributing control across the network, there is no single target for attacks. Additionally, the use of blockchain ensures that data is tamper-proof and verifiable.

Enhanced privacy: Devices can exchange data directly without needing to rely on a third-party cloud platform, reducing the risk of data breaches or unauthorised access.

Scalability: Decentralised networks can scale more efficiently since they don't rely on centralised infrastructure. The addition of more devices simply adds more nodes to the network, rather than overloading a central server.

Why blockchain for IoT?

Blockchain technology offers numerous advantages that directly address the challenges faced by centralised IoT networks.

Data integrity and privacy: IoT devices generate vast amounts of sensitive data, from health information to environmental sensors. Blockchain ensures that this data is immutable and secure by creating a tamper-proof record of all transactions. Every data exchange between IoT devices is verified, reducing the risk of data manipulation and ensuring privacy.

Security: Centralised IoT systems are often vulnerable to cyberattacks. Hackers targeting the central server can gain access to the entire network. Blockchain-based decentralised systems, on the other hand, eliminate this risk by distributing control across all network participants. Each device in the network validates transactions, ensuring the system's resilience against attacks.

Trust and transparency: Blockchain eliminates the need for a trusted third party by allowing devices to communicate and share data securely in a trustless environment. Transactions are recorded on a public ledger that is accessible to all participants, ensuring complete transparency and enabling participants to verify data integrity without relying on any central authority.

Decentralised control: In a decentralised IoT system, devices communicate directly with each other, without the need for a centralised server to manage communication. This peer-to-peer architecture reduces latency and improves real-time decision-making, which is essential in many IoT applications, such as autonomous vehicles or smart grids.

IOTA: Enabling scalable and secure IoT networks

IOTA is one of the most prominent projects in the decentralised IoT space. Unlike traditional blockchain platforms, IOTA uses a unique structure called the Tangle, a directed acyclic graph (DAG). This architecture enables IOTA to scale efficiently and avoid some of the common limitations of blockchain, such as high transaction fees and low throughput. The key features of IOTA are:

Scalability: The Tangle architecture allows for fee-less transactions and high scalability, making it ideal for IoT applications where devices communicate frequently and exchange small amounts of data.

Secure data exchange: IOTA ensures secure communication between devices by storing data on the Tangle, making it tamper-proof and verifiable.

Microtransactions: It enables secure and efficient microtransactions, which are vital for IoT ecosystems where devices often exchange small amounts of data or value.

Zero fees: IOTA eliminates transaction fees, a critical feature for IoT networks that need to handle frequent, small data exchanges.

Here are a few use cases of IOTA.

Smart cities: IOTA's fee-less transactions and scalability make it a great fit for smart city applications, such as smart parking, environmental monitoring, and energy management.

Supply chain management: Its ability to track goods across the supply chain and securely share data in real time makes it ideal for logistics and inventory management.

Autonomous vehicles: IOTA's scalable and secure network enables autonomous vehicles to communicate with each other and exchange data, improving traffic management and safety.

Streamr: Real-time data stream processing for IoT

Streamr is a decentralised data platform that focuses on real-time data streams, enabling IoT devices to share and process data in real time. By leveraging blockchain, Streamr ensures the data is authentic, and that its ownership and privacy are maintained throughout its exchange. Its key features are:

Real-time data processing: Streamr allows IoT devices to transmit data in real time, reducing latency and enabling faster decision-making.

Decentralised data marketplace: Its platform includes a decentralised marketplace where users can buy and sell data streams. This opens up new opportunities for monetising IoT data while ensuring data privacy.

Blockchain integration: Streamr integrates blockchain technology to ensure transparency and authenticity in the